

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		No unwanted bacteria (from forceps) / avoid contamination/ kills bacteria (on forceps)		1		1		1
		(ii)		{Close/ shut} the petri dish and {tape /seal}/ put the lid on and {tape /seal}/ cover and {tape /seal}		1		1		1
	(b)			25°C	1			1		1
	(c)	(i)		380 = 3 marks If incorrect award 2 marks for 381 / 379.9 (rounded incorrectly) 379.94 (not rounded) 380.1327... (using π from calculator) Award 1 mark for 121 x 3.14/ π , seen anywhere in the working Allow answers which are not written in the table.		3		3	3	3
		(ii)	I. II.	Scale on y axis (1) Three bars with correct heights (+/- one small square) and labelled = 2 marks Two bars correct and labelled – 1 mark Allow ecf from the table Ignore any uneven width of bars		1 2		3	1 2	3

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		Antibiotic D (1) {Largest/ biggest} clear area (1) where bacteria have been {killed/ destroyed/ have been prevented from growing} (1) Allow ecf from bar chart.			1 1 1	3		3
	(d)	(i)		Ask another {student/ scientist} to repeat the (same) experiment / compare results with another {group/ scientist} reject repeat unqualified			1	1		1
		(ii)		{Wider range/different} antibiotics / different (types of) bacteria/ different concentrations of antibiotics / different incubation temperatures			1	1		1
				Total for question 6	1	8	5	14	6	14